



CAN-BUS-BASED DOPPLER RADAR And IMU Sensor System

ASHWINI KUMAR SINHA

The controller area network (CAN) bus is a reliable, real-time communication protocol and remains the backbone of modern automotive, electric vehicle, aerospace, and marine systems. It enables the seamless integration of sensors into existing systems by facilitating communication between sensors, actuators, and electronic control units (ECUs) using just two wires.

Advanced driver-assistance systems (ADAS), lidar, and radar sensors often need to coexist on the same CAN network without increasing system complexity or cost. To address this challenge, this CAN-bus-based Doppler radar and IMU (inertial measurement unit) sensor system is designed for easy integration into existing vehicle architectures.

The system combines an HB100 Doppler radar with an LSM303AGR IMU and transmits all critical data

over the vehicle's native CAN bus using standard 11-bit identifiers (0x100, 0x200, and 0x300). This direct compatibility eliminates the need for gateways or protocol conversions, ensuring seamless integration with other ECUs.

The Doppler radar accurately measures the relative speed between the host vehicle and surrounding objects using the Doppler effect. This relative velocity data is crucial for ADAS applications such as collision avoidance, adaptive cruise control, and autonomous braking. While many such systems are laser-based, they may not perform reliably in foggy or very low-light conditions. In contrast, the Doppler radar operates effectively in all lighting conditions, including fog, and is capable of measuring very high speeds that many other systems cannot.

The integrated IMU provides

real-time acceleration and orientation data (yaw, roll, and pitch) and detects critical events, including sudden acceleration, braking, collisions, rollovers, and free fall. All sensor data, including Doppler frequency, relative speed, acceleration, temperature, orientation, and event flags, is transmitted in real time over the CAN bus, enabling other ECUs and controllers to instantly use the data for decision-making.

BILL OF MATERIALS

Name	Designator	Quantity
IndusBoard coin	U1, U5	2
HB100 radar	S1	1
LM358 amplifier	U3	1
CAN bus TJA1050 module	U2, U6	2
OBD port connector	U4	1
Twisted connector wires		1
3D printed enclosure		1
USB-C cable		1

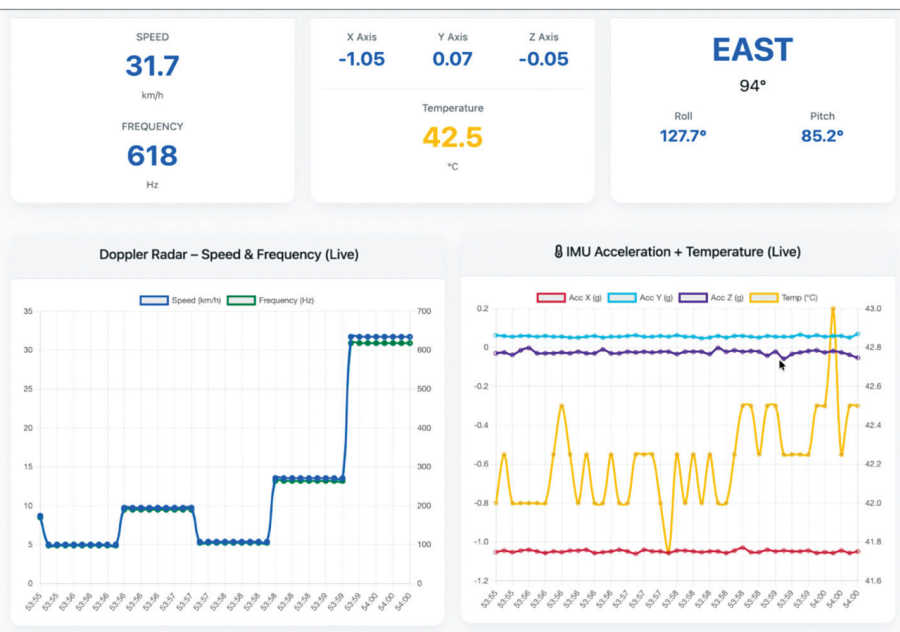
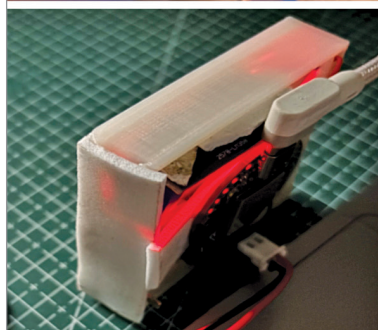


Fig. 1: CAN bus doppler radar and real-time IoT dashboard